

COURSE SYLLABUS

# Querying Microsoft SQL Server 2025

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COURSE NUMBER

**55317B**

DURATION

**3 days · 10 modules**

AUDIENCE LEVEL

**Beginner**

FORMAT

**Instructor-Led / Self-Paced**

# Course syllabus

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## COURSE OVERVIEW

This three-day beginner course teaches students how to use Transact-SQL and other languages to query Microsoft SQL Server 2025 and Azure SQL Database. Using demonstrations and hands-on labs, students will master core querying skills, performance optimization, and semi-structured or vector data handling. This course serves as the foundation for all SQL Server-related disciplines including Database Administration, Database Development, and Business Intelligence.

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## AT COURSE COMPLETION

After completing this course, you will be able to:

Query, sort, filter, and modify data in SQL Server 2025 using Transact-SQL statements.

Query external data sources using Linked Servers, PolyBase, and modern integration techniques.

Programmatically query SQL Server 2025 using stored procedures, functions, and programmatic blocks.

Optimize query performance using indexes, execution plans, and modern execution features.

Work with XML, JSON, and vector-based AI data structures within modern SQL Server environments.

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## PREREQUISITES

- Basic understanding of relational database concepts
  - Familiarity with the Windows operating system environment
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## COURSE OUTLINE

### MODULE 1 Querying Data with SQL Server Management Studio (SSMS)

80 min

Learn to navigate and utilize the primary management interface for SQL Server 2025. This module focuses on executing queries, analyzing results, and customizing the SSMS 22 environment.

#### LESSONS

- Lesson 1:** Introducing SQL Server Management Studio (SSMS) 22
- Lesson 2:** Writing and Executing Queries in SSMS

#### LAB EXERCISES

- Navigating SSMS and Executing Database Queries

#### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Navigate the SQL Server Management Studio (SSMS) 22 environment efficiently.
- Connect to local and cloud-based SQL Server 2025 databases.
- Write, execute, and troubleshoot basic Transact-SQL queries using SSMS tools.
- Query, sort and filter data in database tables in SQL Server 2025

### MODULE 2 Querying Data with Visual Studio Code

80 min

Learn to configure and leverage Visual Studio Code as a modern lightweight editor for querying SQL Server 2025. This module covers essential extensions, connection profiles, and environment operations.

#### LESSONS

- Lesson 1:** Executing T-SQL Queries in Visual Studio Code
- Lesson 2:** Manage Connection Profiles For Local SQL Server 2025 Application

#### LAB EXERCISES

- Querying SQL Server using Visual Studio Code

#### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure Visual Studio Code with SQL Server extensions for query development.
- Manage connection profiles for local SQL Server 2025 instances.
- Write and execute Transact-SQL statements using VS Code shortcuts and features.
- Query, sort and filter data in database tables in SQL Server 2025

### MODULE 3 Basic Transact-SQL Queries

100 min

Master the structural basics of the SELECT statement. This module teaches how to retrieve, filter, sort, and handle null values inside standard relational tables.

#### LESSONS

- Lesson 1:** The Anatomy of a SELECT Statement
- Lesson 2:** Filtering and Sorting Query Results

#### LAB EXERCISES

Retrieving, Filtering, and Sorting Customer Data

#### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Construct valid SELECT statements to retrieve columns from database tables.
- Filter query results using advanced WHERE clause operators and conditions.
- Sort query results chronologically, numerically, or alphabetically.
- Handle NULL values safely in queries using conditional expressions.

### MODULE 4 Advanced Transact-SQL Queries

100 min

Progress to advanced querying concepts including multi-table joins, complex aggregations, subqueries, and modern CTE structures.

#### LESSONS

- Lesson 1:** Combining Datasets with INNER and OUTER JOINS
- Lesson 2:** Aggregating Data using GROUP BY and HAVING
- Lesson 3:** Utilizing Subqueries and Common Table Expressions (CTEs)

#### LAB EXERCISES

Writing Advanced Queries with Joins, Aggregations, Subqueries, and CTEs

#### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Write multi-table queries utilizing INNER, LEFT, RIGHT, and FULL JOIN commands.
- Perform data aggregations using GROUP BY, HAVING, and aggregate functions.
- Construct self-contained and correlated subqueries to filter data dynamically.
- Design clear and readable queries using Common Table Expressions (CTEs).

## MODULE 5 Optimizing Query Performance

130 min

Discover performance tuning tools and methodologies in SQL Server 2025. This module focuses on indexing mechanics, execution plan analysis, and identifying database bottlenecks.

### LESSONS

- Lesson 1:** Understanding SQL Server Indexing Concepts
- Lesson 2:** Analyzing Execution Plans in SSMS
- Lesson 3:** Query Performance Tuning Best Practices

### LAB EXERCISES

- Identifying and Resolving Query Bottlenecks

### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Explain the performance impact of Clustered and Nonclustered indexes.
- Analyze execution plans to identify table scans, index scans, and index seeks.
- Refactor slow-performing queries to optimize processing times.
- Query, sort and filter data in database tables in SQL Server 2025

## MODULE 6 Modifying Data with INSERT, UPDATE, and DELETE

110 min

Understand core mechanisms of Data Manipulation Language (DML). This module covers inserting rows, modifying values, removing data, and utilizing transaction controls to protect data integrity.

### LESSONS

- Lesson 1:** Inserting and Updating Database Records
- Lesson 2:** Deleting Data safely and Using Transactions

### LAB EXERCISES

- Performing Safe Data Manipulation Operations

### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Execute INSERT statements to append single and multiple rows into tables.
- Perform UPDATE operations targeting specific columns and rows.
- Run DELETE queries safely using filtering predicates and transaction rollbacks.
- Control transaction scopes using COMMIT and ROLLBACK commands.

## MODULE 7 Working with XML, JSON, and Vector (AI) Data

120 min

Query and process complex, modern data structures inside SQL Server 2025. This module covers semi-structured formatting with JSON/XML, and introduces vector representations for AI applications.

### LESSONS

- Lesson 1:** Querying and Formatting JSON Data in SQL Server 2025
- Lesson 2:** Handling XML Data Structures
- Lesson 3:** Introduction to Vector Data Types and AI-Ready Queries

### LAB EXERCISES

Querying and Processing JSON, XML, and Vector Data in SQL Server 2025

### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Format and retrieve database data to and from JSON structures.
- Query semi-structured XML fields using XQuery elements.
- Describe the significance of the vector data type and AI queries in SQL Server 2025.
- Query, sort and filter data in database tables in SQL Server 2025

## MODULE 8 Using Stored Procedures and Functions

120 min

Encapsulate querying logic inside reusable database modules. This module highlights stored procedure patterns, input/output variables, and the creation of deterministic functions.

### LESSONS

- Lesson 1:** Creating and Executing Stored Procedures with Parameters
- Lesson 2:** Implementing Scalar and Table-Valued Functions

### LAB EXERCISES

Building Stored Procedures and Custom Functions

### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and execute stored procedures featuring default and customized parameters.
- Implement custom Scalar functions to process singular return values.
- Write and optimize Table-Valued functions for complex inline lookups.
- Query, sort and filter data in database tables in SQL Server 2025

## MODULE 9 Querying SQL Server Programmatically

110 min

Utilize programmatic controls in Transact-SQL. This module covers variable declarations, iterative logic loops, branching parameters, and robust structured error handling mechanisms.

### LESSONS

- Lesson 1:** Implementing Procedural Logic with T-SQL Blocks
- Lesson 2:** Handling Execution Errors and Implementing Try-Catch

### LAB EXERCISES

- Writing Programmatic Queries with Error Handling

### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Declare and manipulate local T-SQL variables dynamically.
- Control execution paths using IF-ELSE branching and iterative loops.
- Implement structural error handling routines using TRY-CATCH blocks.
- Query, sort and filter data in database tables in SQL Server 2025

## MODULE 10 Managing External Data Sources

130 min

Query heterogeneous datasets external to SQL Server. This module covers establishing secure Linked Server definitions and leveraging PolyBase to reference external tabular and object storage.

### LESSONS

- Lesson 1:** Querying Remote Databases with Linked Servers
- Lesson 2:** Integrating Heterogeneous Data with PolyBase
- Lesson 3:** Configuring Modern External Data Connections

### LAB EXERCISES

- Querying External Data Sources using PolyBase and Linked Servers

### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure and query remote database engines using Linked Servers.
- Explain the configuration and storage advantages of PolyBase.
- Query distinct external datasets through native virtualization techniques.
- Query, sort and filter data in database tables in SQL Server 2025